All-optical wavelength conversion

11/8/2018

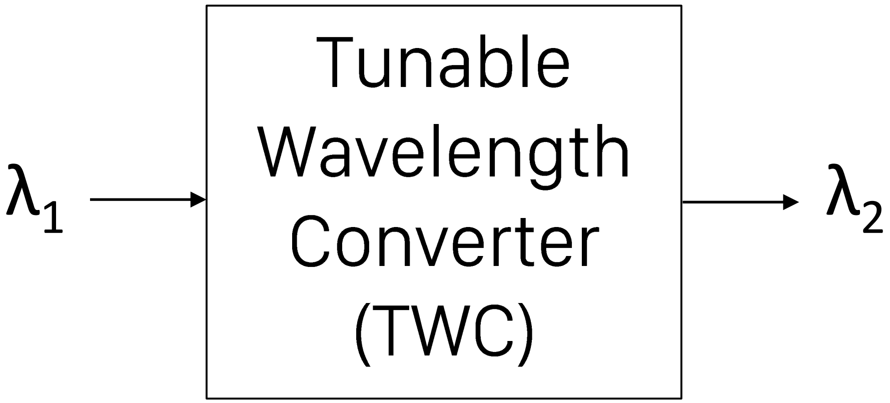
University of Southern California

Team 558 - Optical Fiber Communication Systems

Prof. Willner

Presented by Binh Phan

Motivation



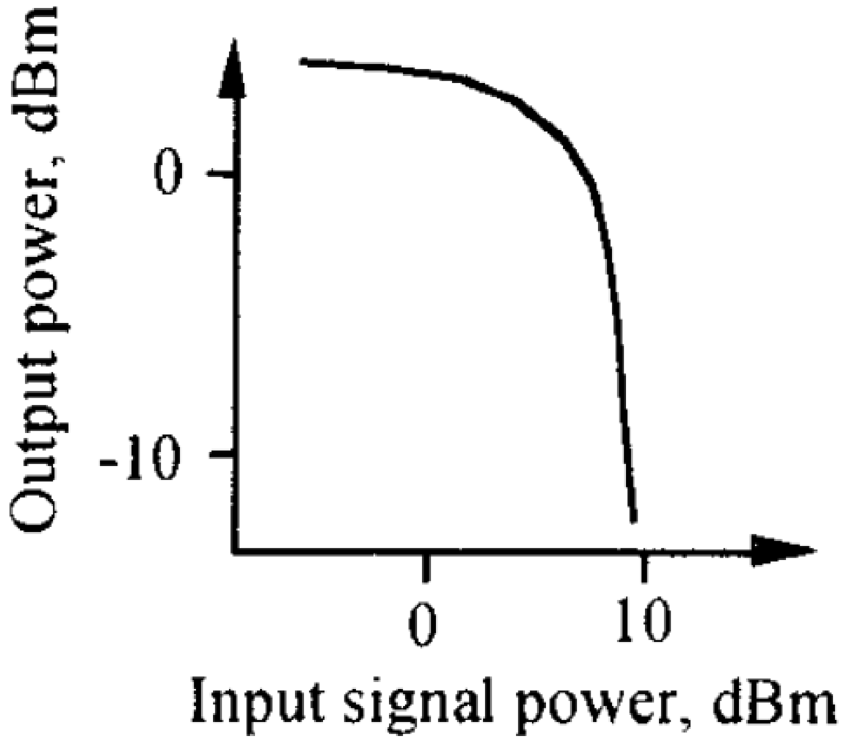
All-optical: no electronics – too slow and costly

Applications: packet switching, contention resolution

Outline

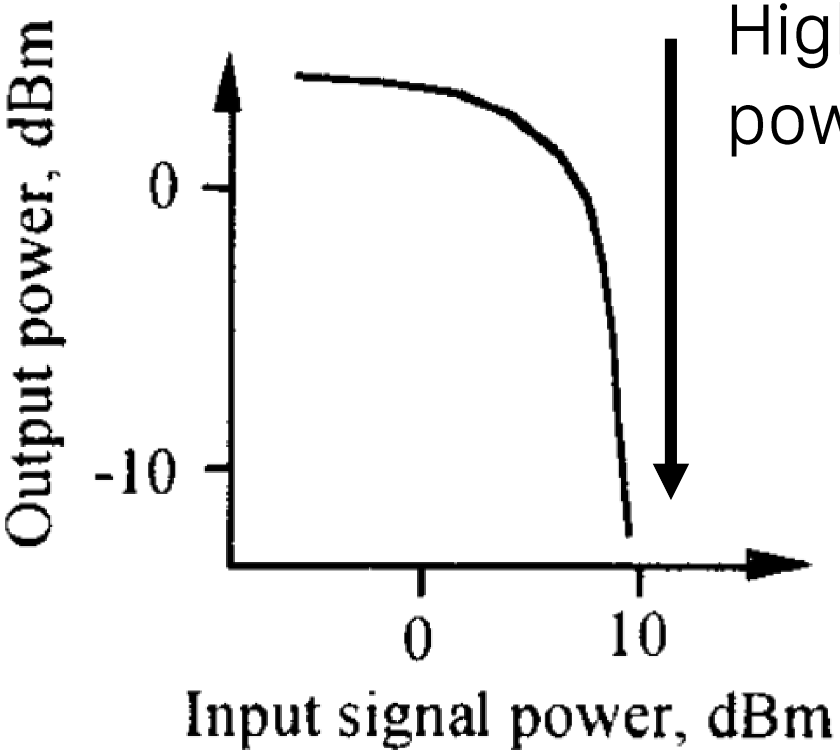
1. Semiconductor Optical Amplifier (SOA)
2. AOWC using cross-gain modulation (XGM) in SOA
3. AOWC using cross-phase modulation (XPM) in SOA
4. AOWC using four-wave mixing (FWM) in SOA
5. Performance Comparison

Semiconductor Optical Amplifier (SOA)



Stubkjær, K., Jørgensen, C., Danielsen, S. L., Mikkelsen, B., Vaa, M., Pedersen, R. J. S., ... Ratovelomanana, F. (1996). Wavelength conversion devices and techniques. In Proceedings of the 22nd European Conference on Optical Communication (Vol. Volume 4, pp. 33–40). IEEE.

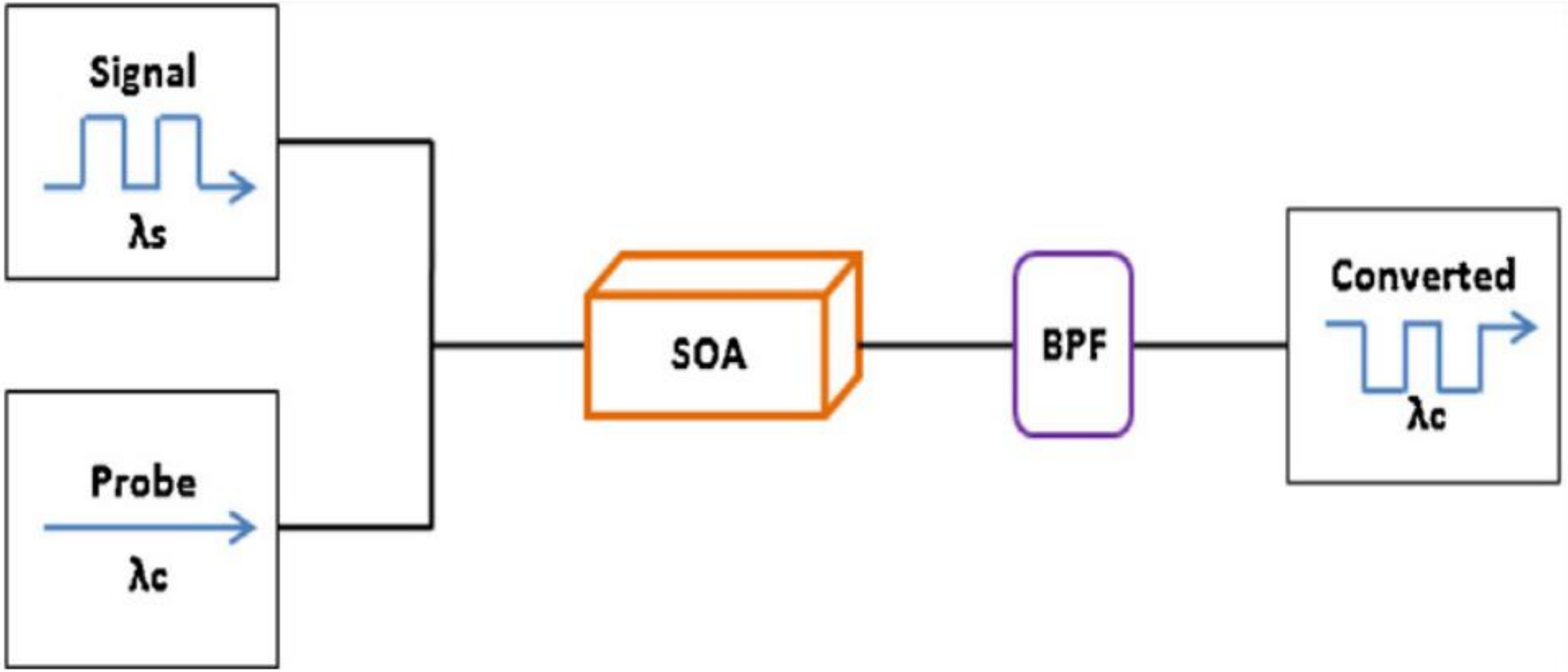
Semiconductor Optical Amplifier (SOA)



High input signal power -> No gain

Stubkjær, K., Jørgensen, C., Danielsen, S. L., Mikkelsen, B., Vaa, M., Pedersen, R. J. S., ... Ratovelomanana, F. (1996). Wavelength conversion devices and techniques. In Proceedings of the 22nd European Conference on Optical Communication (Vol. Volume 4, pp. 33–40). IEEE.

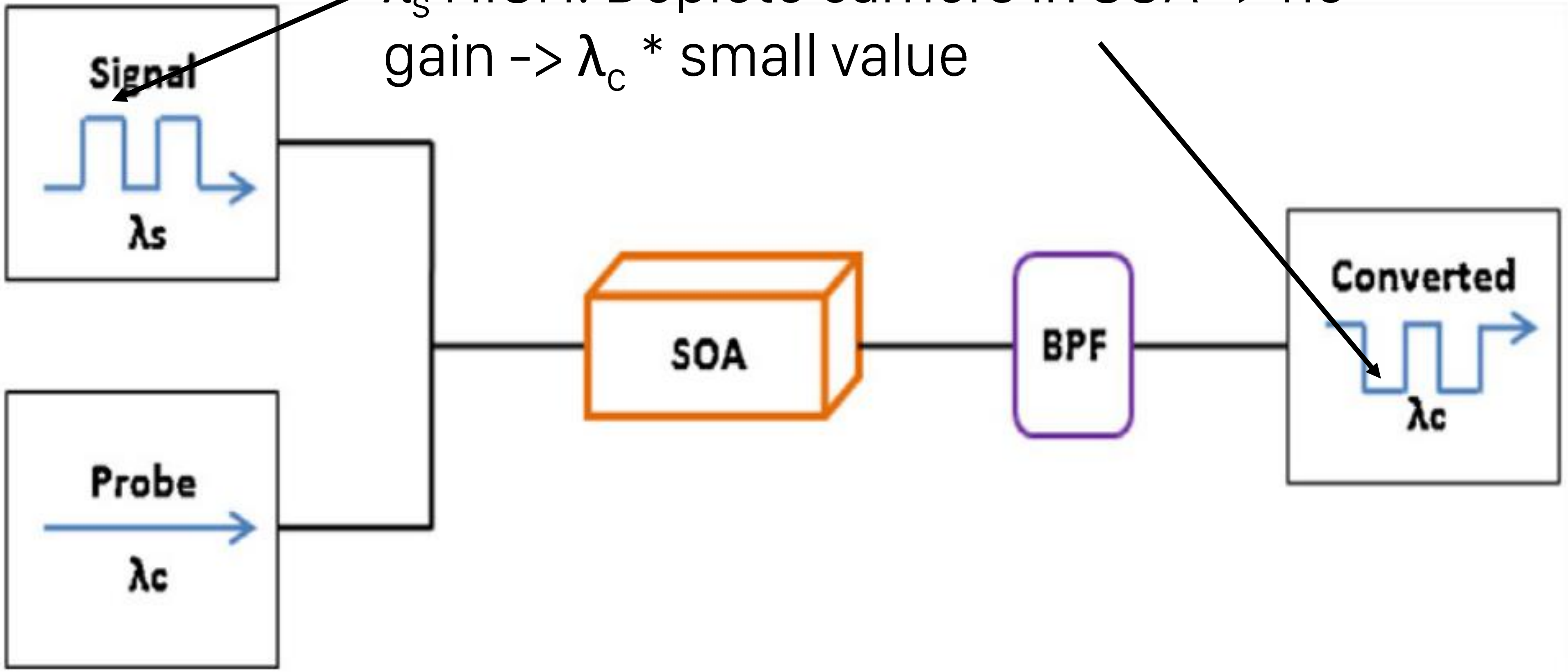
AOWC using cross-gain modulation (XGM) in SOA



Farah Diana Mahad,Abu Sahmah M.Supa'at. "Comparative performance testing of SOA wavelength conversion techniques for future all-optical systems," International Journal for Light and Electron Optics (Optik), April 2012.

AOWC using cross-gain modulation (XGM) in SOA

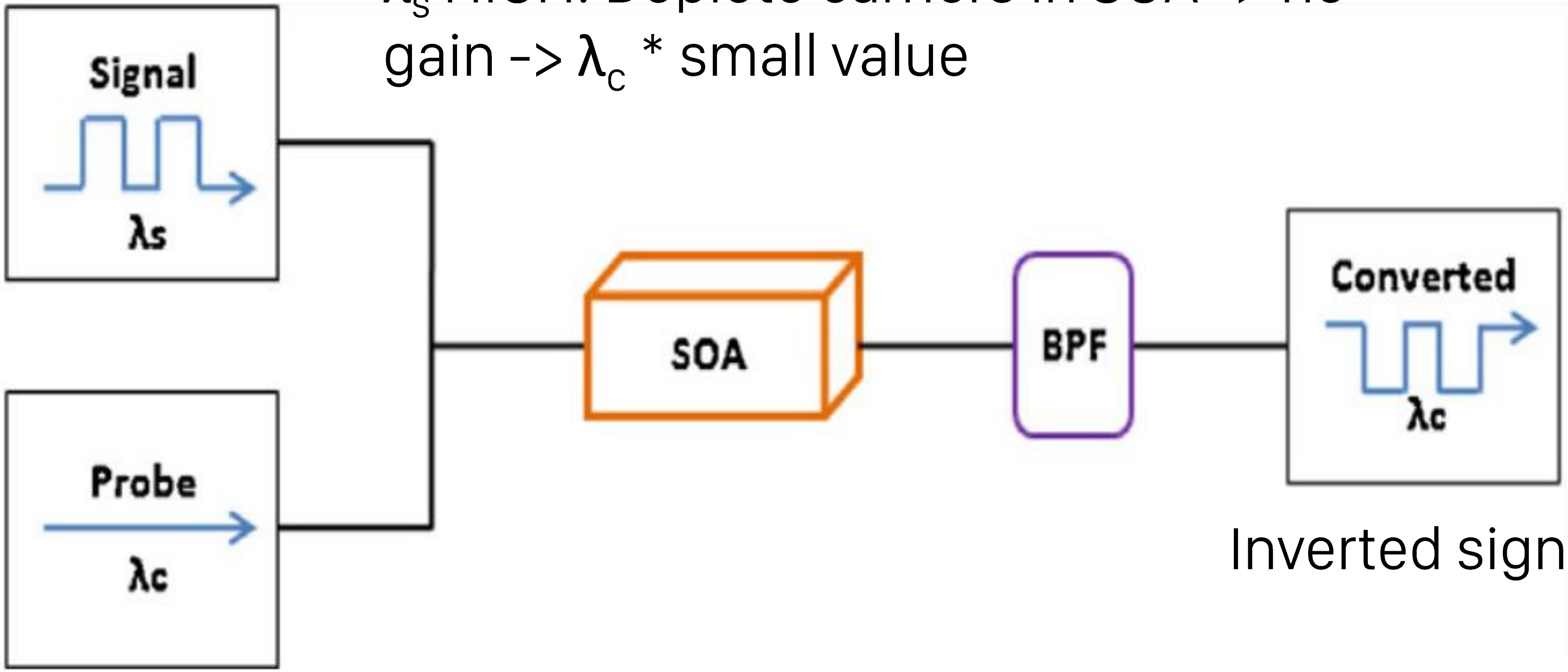
λ_s HIGH: Deplete carriers in SOA -> no gain -> λ_c * small value



Farah Diana Mahad,Abu Sahmah M.Supa'at. "Comparative performance testing of SOA wavelength conversion techniques for future all-optical systems," International Journal for Light and Electron Optics (Optik), April 2012.

AOWC using cross-gain modulation (XGM) in SOA

λ_s HIGH: Deplete carriers in SOA -> no gain -> λ_c * small value

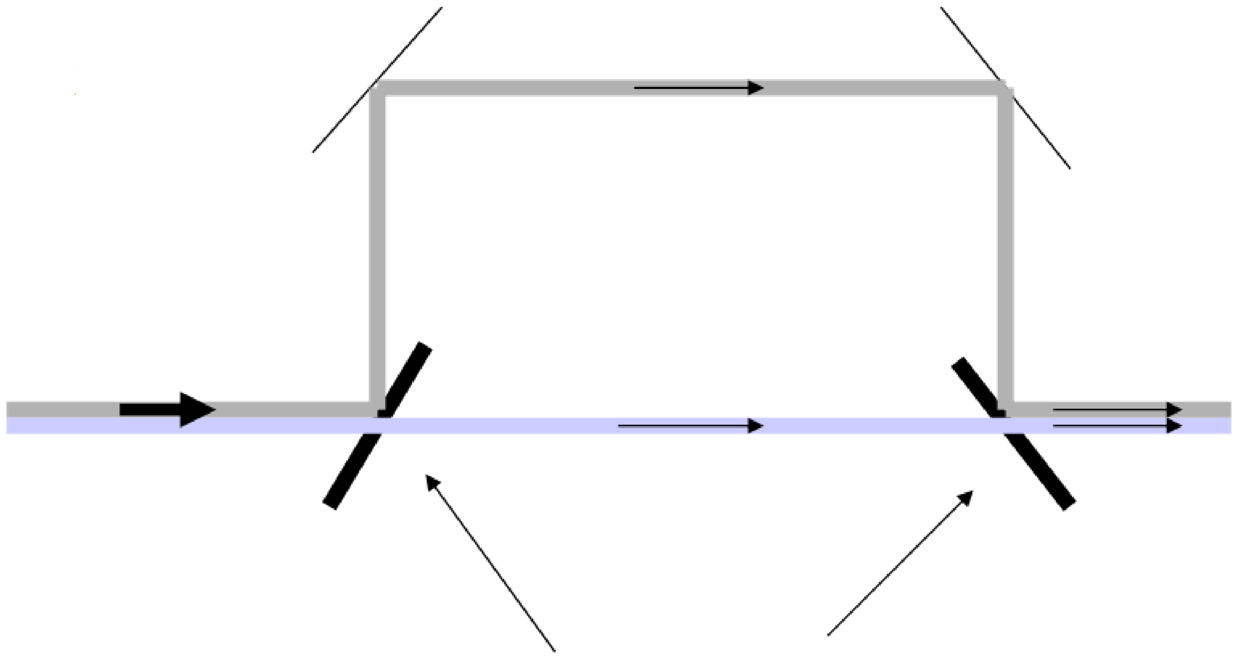


Inverted signal

Farah Diana Mahad,Abu Sahmah M.Supa'at. "Comparative performance testing of SOA wavelength conversion techniques for future all-optical systems," International Journal for Light and Electron Optics (Optik), April 2012.

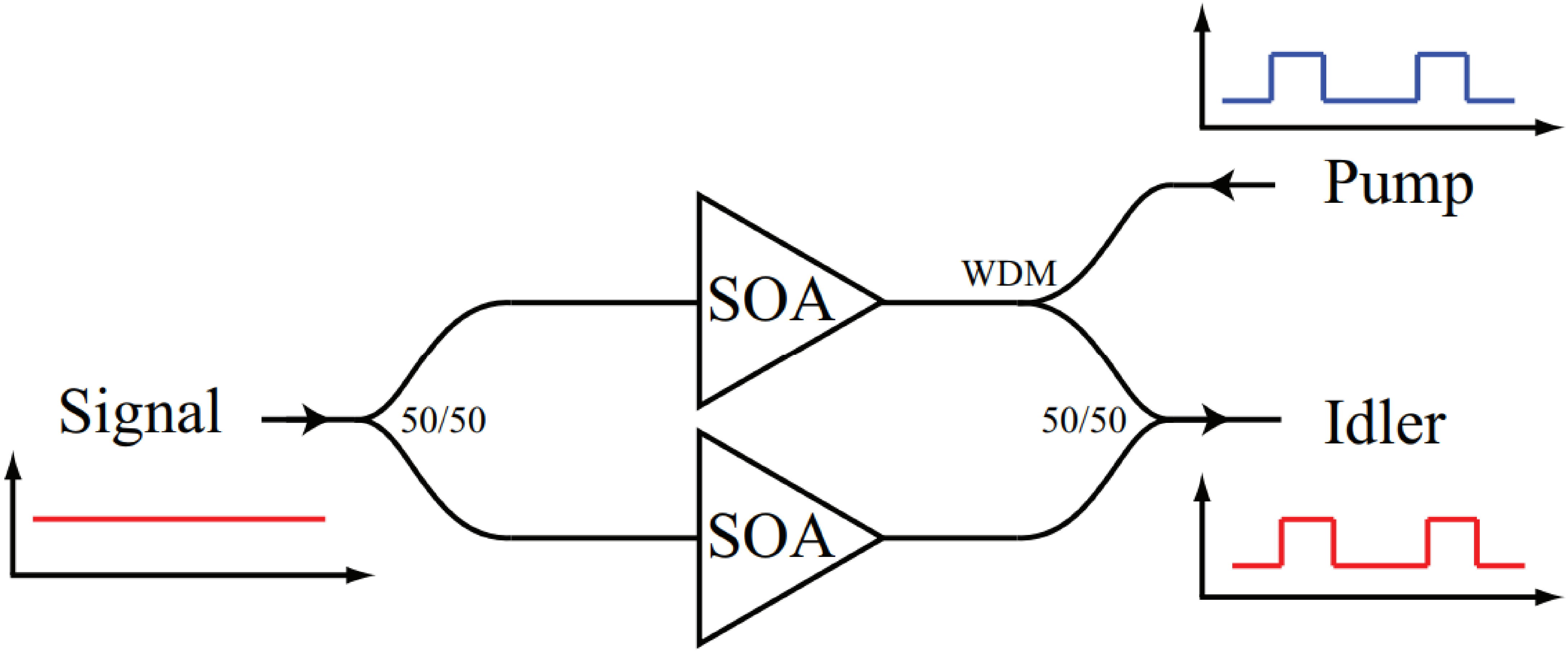
AOWC using cross-phase modulation (XPM) in SOA

Mach-Zender Interferometer



Converts phase modulation to intensity modulation

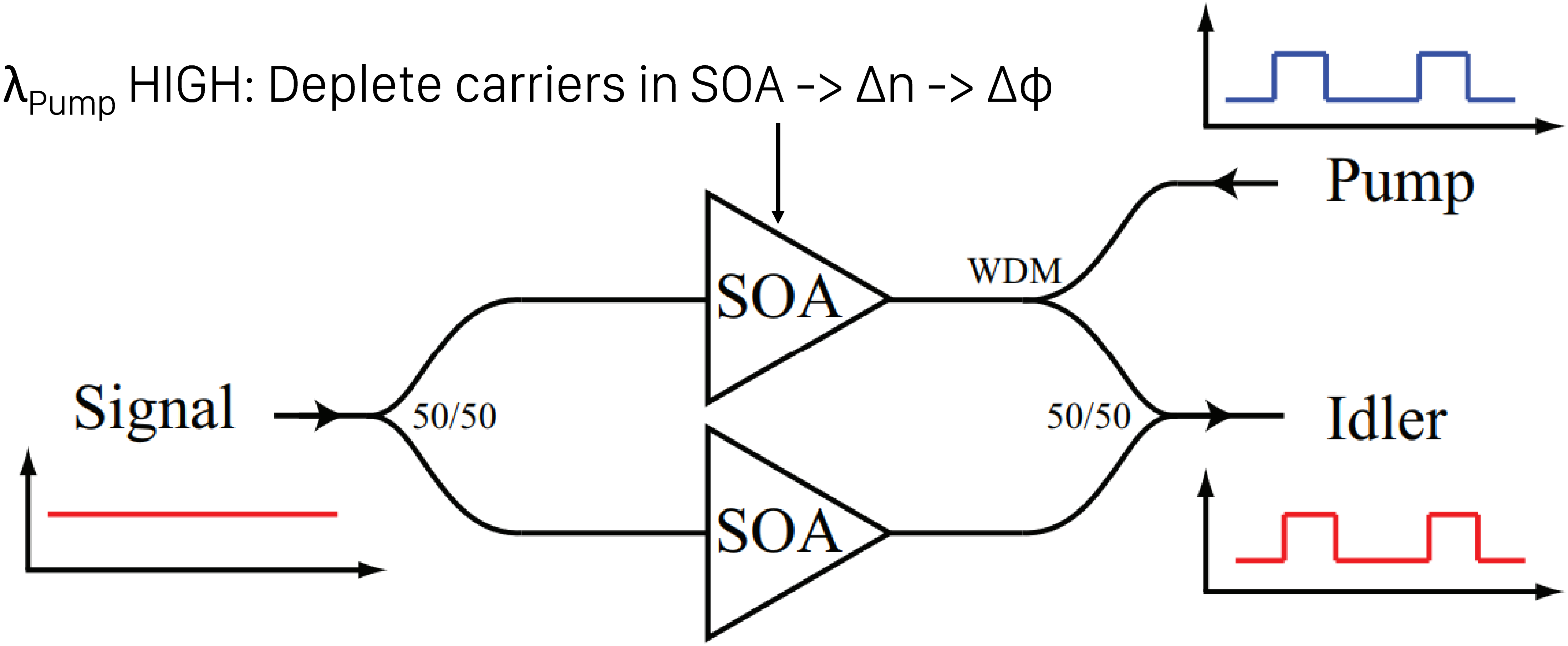
AOWC using cross-phase modulation (XPM) in SOA



Provo, Richard (2010). *Nonlinear Techniques for Optical Wavelength Conversion* (Doctoral dissertation). Retrieved from The University of Auckland

AOWC using cross-phase modulation (XPM) in SOA

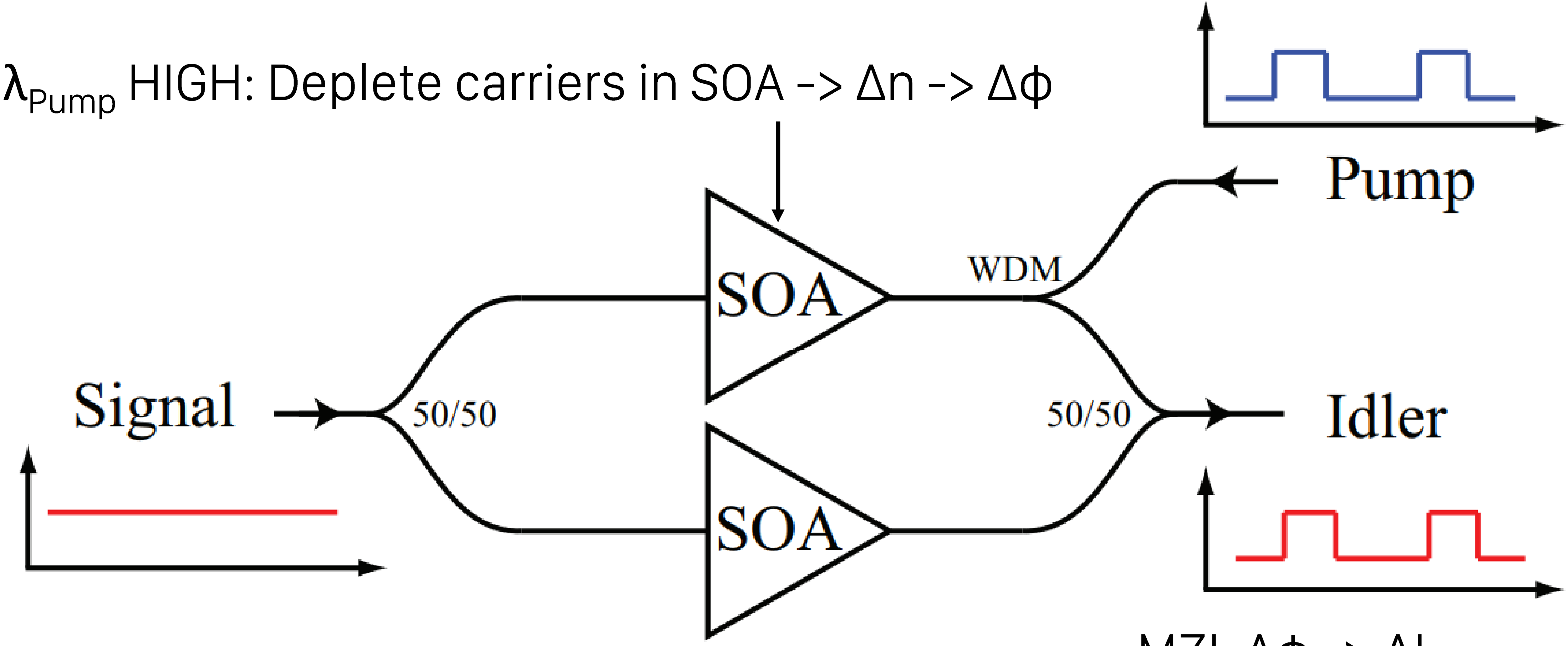
λ_{Pump} HIGH: Deplete carriers in SOA -> Δn -> $\Delta \phi$



Provo, Richard (2010). *Nonlinear Techniques for Optical Wavelength Conversion* (Doctoral dissertation). Retrieved from The University of Auckland

AOWC using cross-phase modulation (XPM) in SOA

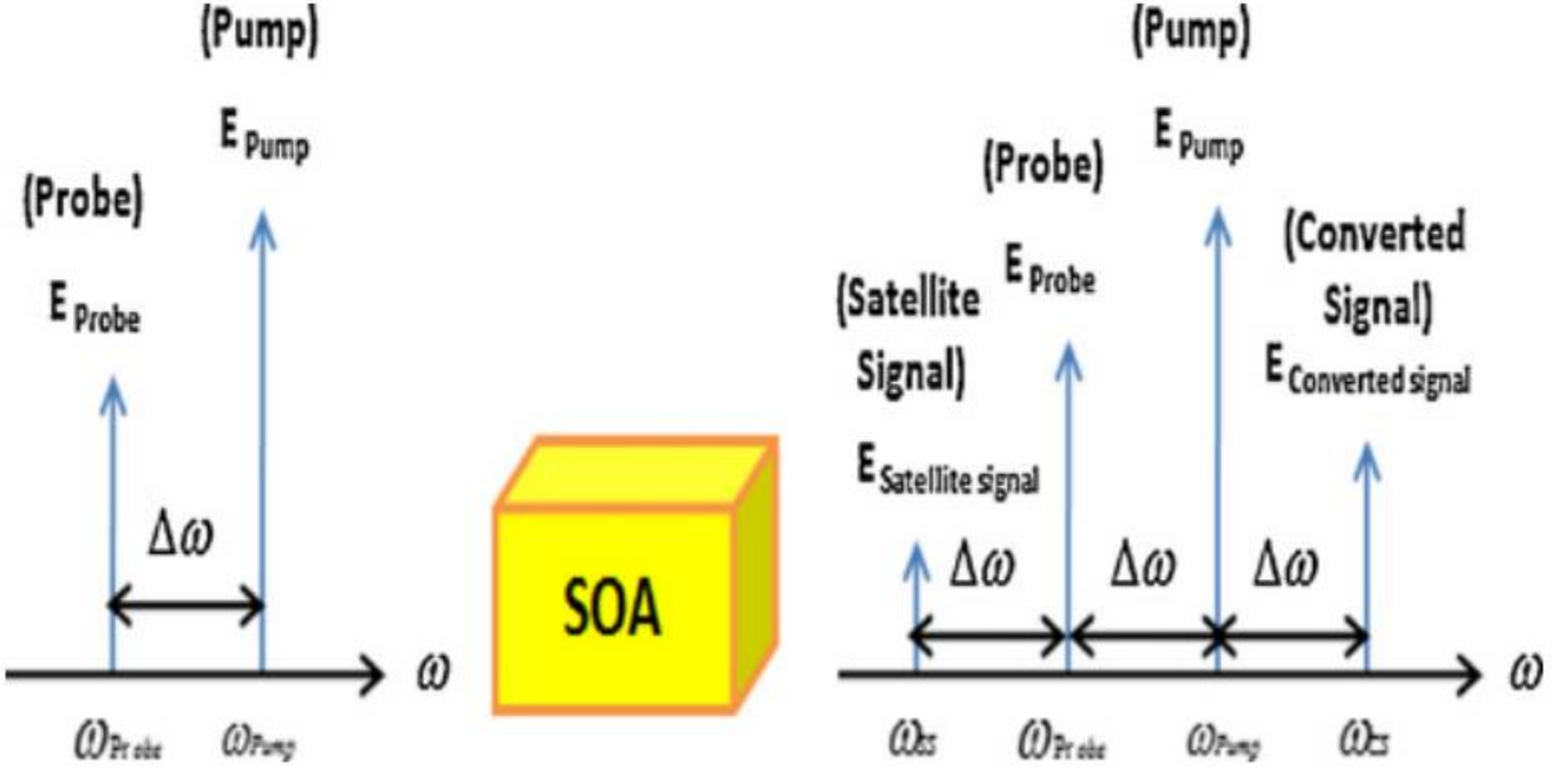
λ_{Pump} HIGH: Deplete carriers in SOA -> Δn -> Δϕ



MZI: Δϕ -> ΔI

Provo, Richard (2010). *Nonlinear Techniques for Optical Wavelength Conversion* (Doctoral dissertation). Retrieved from The University of Auckland

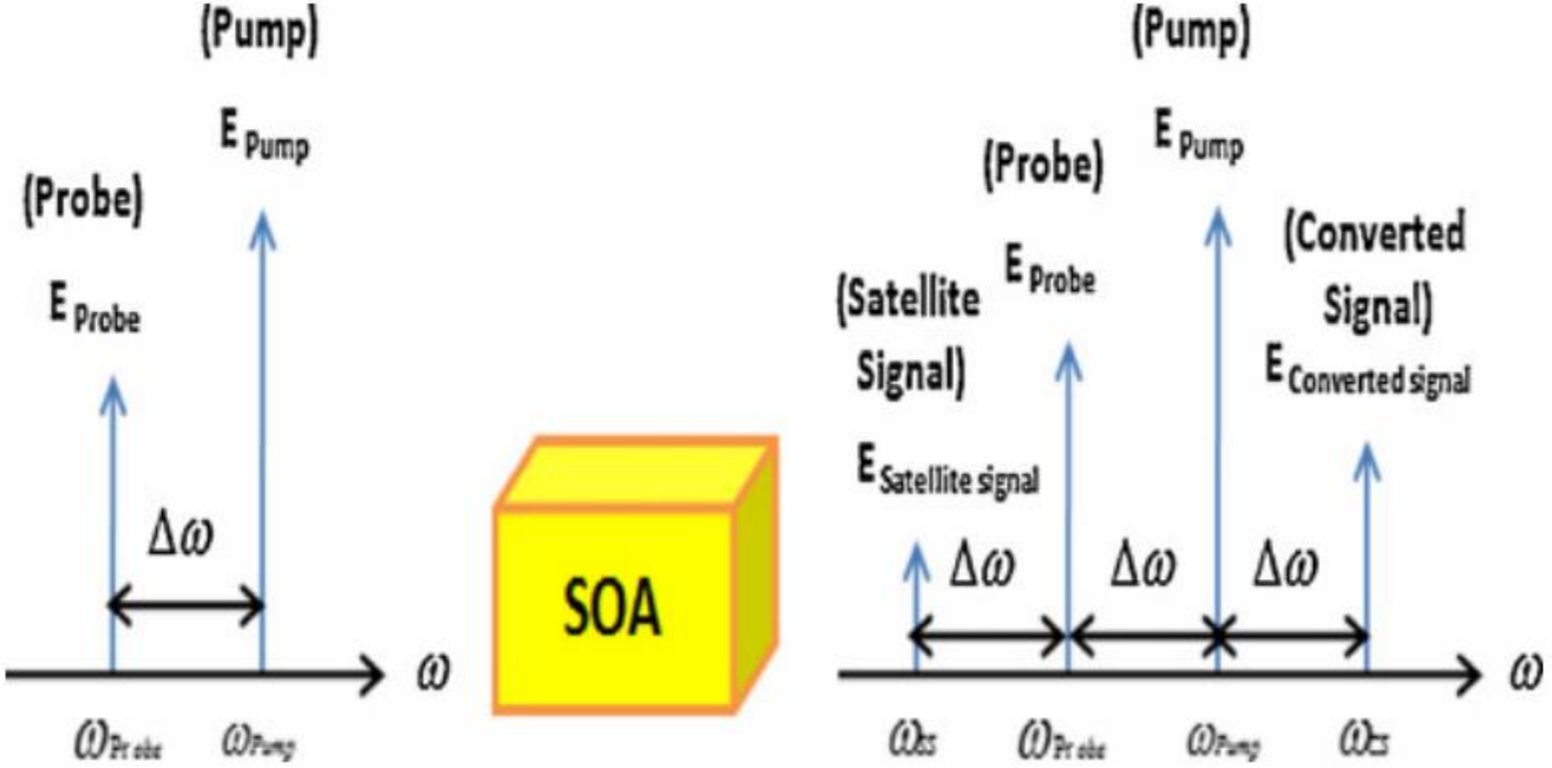
AOWC using four-wave mixing (FWM) in SOA



Farah Diana Mahad,Abu Sahmah M.Supa'at. "Comparative performance testing of SOA wavelength conversion techniques for future all-optical systems," International Journal for Light and Electron Optics (Optik), April 2012.

AOWC using four-wave mixing (FWM) in SOA

$$2\cos\theta\cos\varphi = \cos(\theta-\varphi) + \cos(\theta+\varphi)$$



Farah Diana Mahad,Abu Sahmah M.Supa'at. "Comparative performance testing of SOA wavelength conversion techniques for future all-optical systems," International Journal for Light and Electron Optics (Optik), April 2012.

Performance Comparison

	Supports 10 Gb/s	Supports PSK and other modulation formats
XGM	Yes	No
XPM	Yes	No
FWM	Yes	Yes

Each have their pros and cons.

Summary

- AOWC is essential for packet switching and contention resolution
- XGM, XPM, FWM
- Fast, supports 10 Gb/s
- XGM and XPM only support ASK, FWM can support PSK and other modulation formats

Thank you!